

Business Requirements Document & Solution Architecture

Retail Customer Intelligence Platform

Mini Enterprise Data Platform — turning raw e-commerce transactions into customer segments, churn predictions, and an exportable retention list.

Project	Retail Customer Intelligence Platform
Programme	AIO Conquer 2026 · Module 01 · Project #22 + #20
Document	BRD + Solution Architecture (combined)
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Jira board	ACM1

0. Document Purpose

This document defines **why** the project exists before **how** it is built. Part A (§1–§9) captures the business problem, measurable objectives, stakeholders, scope, and requirements. Part B (§10–§13) captures the solution architecture, data model, and UML design that realise those requirements. It is the single source of truth that downstream artefacts — pipeline, tests, and the final business-impact report — are traced back to.

Guiding principle — SDLC over model. What makes this project "enterprise" is the discipline of the full lifecycle (requirements → design → data modelling → build → testing → CI/CD → impact), not the sophistication of the churn model. The predictive model represents only ~15–20% of total project value.

Part A — Business Requirements

1. Business Context & Problem Statement

The business runs an online retail operation with a large base of historical transactions but **no systematic way to understand or act on customer behaviour**. Marketing and retention activities are currently undifferentiated.

1.1 Pain Points

#	Pain point	Business consequence
P1	Marketing sends mass campaigns to the entire customer base	Low relevance, wasted marketing spend, message fatigue
P2	Retention is low ; high-value customers churn unnoticed	Lost recurring revenue from the most profitable customers
P3	No systematic way to identify which customers are about to leave	Reactive (not proactive) retention; intervention happens too late

1.2 Root Cause

Customer data exists only as raw transaction records. There is no cleaned, modelled, and segmented view of customers — and no risk score — so the business cannot target, prioritise, or measure retention efforts.

2. Business Objectives & Success Metrics

The platform must enable measurable improvements in retention and campaign efficiency.

Goal	Target	Metric	Owner
Reduce churn	−10%	Churn rate (rolling 90-day)	Marketing
Grow retention revenue	+5%	Revenue from returning customers	Marketing
Improve campaign efficiency	Targeted, not mass	% of campaigns sent to defined segments	Marketing

These are **business outcome targets** the platform is designed to support. The platform's own delivery success is defined separately in §7 (Acceptance Criteria).

3. Stakeholders

Stakeholder	Role / Interest	Primary need from the platform
Marketing Manager	Owns campaigns & retention	Targeted retention lists, clear segment definitions, exportable lists
CRM Team	Executes customer engagement	Customer-level churn risk, an export workflow they can action
Data Team	Builds & operates the platform	Reliable pipeline, monitored model performance, data quality visibility
Project Sponsor / Reviewer	Evaluates delivery	Evidence of SDLC maturity and measurable business impact

4. Scope

4.1 In Scope

The end-to-end flow:



1. **Ingest & clean** historical e-commerce transactions into a governed, layered data store.
2. **Model** the data into a dimensional (star) schema with measures and conformed dimensions.
3. **Compute RFM** (Recency, Frequency, Monetary) scores and assign business segment labels (Champions, Loyal, At Risk, Lost, ...).
4. **Predict churn risk** at the customer level (batch scoring).
5. **Serve** segments, KPIs, churn-risk, and cohort views through a dashboard.
6. **Export** a retention list (e.g. At-Risk + high-value customers) for Marketing.
7. **Monitor** data quality and model performance.

4.2 Out of Scope

The following are explicitly **excluded** from this delivery (simulated or documented only):

- Live campaign execution.
- Email / SMS sending.
- Real CRM-system integration (interaction is simulated via list export).
- Real-time / streaming scoring (a streaming replay is a documented future option only).
- Cloud production deployment (a target architecture is documented; the build is local/containerised).

4.3 Assumptions

- The source is a single historical transaction dataset (UCI Online Retail II, ~1.07M rows, CC BY 4.0).
- Source timestamps are historical (2009–2011) and are **date-rebased to the present** so that recency and churn windows remain meaningful for demonstration.
- One currency (GBP); monetary values are reported in GBP.
- Customer identity is the source-provided **Customer ID**; transactions without a customer ID are treated as guest/anonymous and excluded from customer-level analytics.

4.4 Constraints

- **Timeline:** 4-week delivery window (4 sprints).
- **Team:** 5 members (see §14).
- **Technology discipline:** deliberately **no Kafka / Spark / Kubernetes / microservices** — maturity is demonstrated through SDLC completeness, not framework count.
- **Reproducibility:** the stack must run locally with a single command for reviewer reproducibility.

5. Functional Requirements

ID	Requirement	Priority	Traces to
FR-1	The system shall ingest raw transaction records into an immutable raw layer.	MUST	P2, P3

ID	Requirement	Priority	Traces to
FR-2	The system shall clean, deduplicate, type-cast, and validate transactions, and quarantine batches that fail quality checks.	MUST	P2
FR-3	The system shall rebase historical transaction dates to the present so recency/churn windows are meaningful.	MUST	P3
FR-4	The system shall model transactions into a star schema (<code>fact_transactions</code> + <code>dim_customer</code> , <code>dim_product</code> , <code>dim_date</code> , <code>dim_country</code>).	MUST	P1
FR-5	The system shall compute per-customer RFM scores (1–5 quintiles) and assign a segment label.	MUST	P1
FR-6	The system shall produce a per-customer churn probability and a threshold-based churn flag.	SHOULD	P2, P3
FR-7	The system shall present segments, KPIs, churn-risk, and cohort-retention views in a dashboard.	MUST	P1
FR-8	The system shall export a retention list (filterable by segment / churn risk) for Marketing.	MUST	P1, P2
FR-9	The system shall monitor data quality (row counts, null rates, freshness) and model performance (drift, AUC) each run.	SHOULD	P2
FR-10	Pipeline runs shall be orchestrated and schedulable.	SHOULD	P3

Priority uses MoSCoW. MUST = required for the credible vertical-slice demo; SHOULD = high value, included if time permits; remaining items are future roadmap.

6. Non-Functional Requirements

ID	Category	Requirement
NFR-1	Data quality	Key columns (customer/invoice/product) validated for uniqueness, completeness, and referential integrity; failures gated, not silently passed.
NFR-2	Reproducibility	Entire stack runs from a single command in a containerised environment; no manual environment drift.
NFR-3	Reliability	Ingestion is idempotent and partitioned; a failed run does not corrupt prior data; the raw layer is immutable.
NFR-4	Auditability	Each pipeline stage writes an audit/quality log (row counts, metrics, timestamps).
NFR-5	Performance	The full historical dataset (~1.07M rows) processes end-to-end within the local demo window.

ID	Category	Requirement
NFR-6	Maintainability	Code is tested (unit + pipeline + acceptance) and linted; transformations are versioned.
NFR-7	Model quality	The churn model must meet a minimum performance gate (AUC threshold) before its scores are published.
NFR-8	Usability	Marketing can obtain a retention list without engineering help (self-serve export).

7. Acceptance Criteria (Definition of Success)

The platform delivery is considered successful when **all** of the following hold:

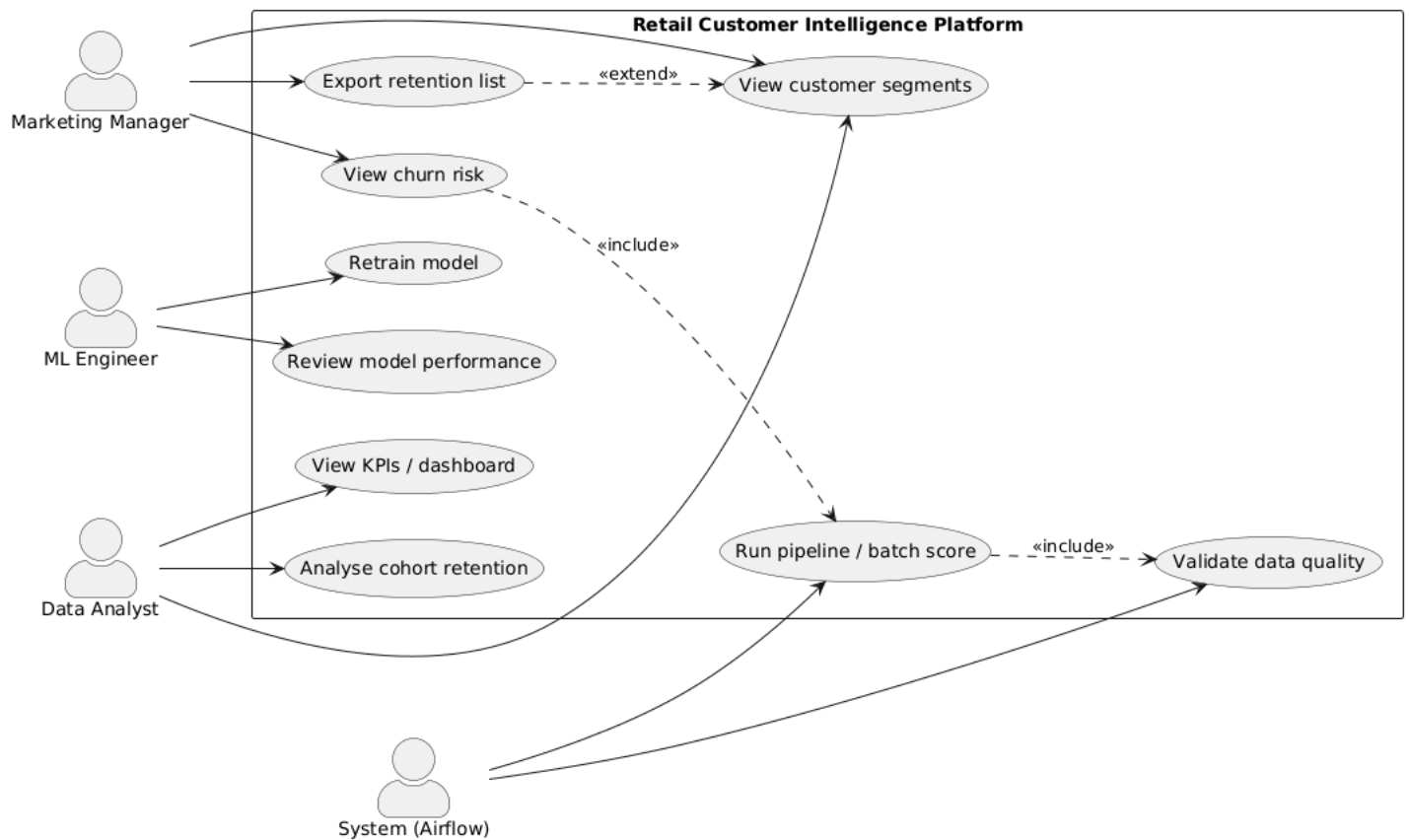
1. The dashboard delivers a **segment view and a churn-risk view**.
2. A **retention list is exportable** by Marketing without manual data work.
3. **Model AUC and data quality are monitored** and visible.
4. **RFM totals reconcile** with raw revenue; **segment counts sum** to the customer base; **dashboard numbers match** the Gold tables.
5. The stack is **reproducible** by a reviewer with a single command.

8. Risks & Mitigations

ID	Risk	Likelihood	Impact	Mitigation
R1	Scope creep — over-investing in ML/tooling vs the SDLC slice	High	High	Enforce MoSCoW; lock the vertical slice; model is last (only ~15–20% of value)
R2	Source data quality issues (cancellations, missing customer IDs, zero/negative values, encoding)	High	Medium	Explicit cleaning rules + quality gate + quarantine; documented in data dictionary
R3	Source file truncated / incomplete (e.g. spreadsheet row limit)	Medium	Medium	Validate row counts and date coverage against the published dataset before sign-off
R4	Date-rebasing logic incorrect → wrong recency/churn windows	Medium	High	Unit tests on date-shift; reconcile shifted vs original dates
R5	Documentation describes technology not yet built	Medium	Medium	Keep BRD/architecture in sync with the as-built repo each sprint
R6	4-week timeline with 5 members	Medium	Medium	Phased build order; clear ownership matrix; weekly demos

9. Use Case Model

The actors and their interactions with the platform.



Actor	Use cases
Marketing Manager	View customer segments · View churn risk · Export retention list
Data Analyst	View customer segments · View KPIs / dashboard · Analyse cohort retention
ML Engineer	Retrain model · Review model performance
System (Airflow)	Run pipeline / batch score · Validate data quality

Key relationships: *View churn risk* «include» *Run pipeline / batch score* «include» *Validate data quality*; *Export retention list* «extend» *View customer segments*.

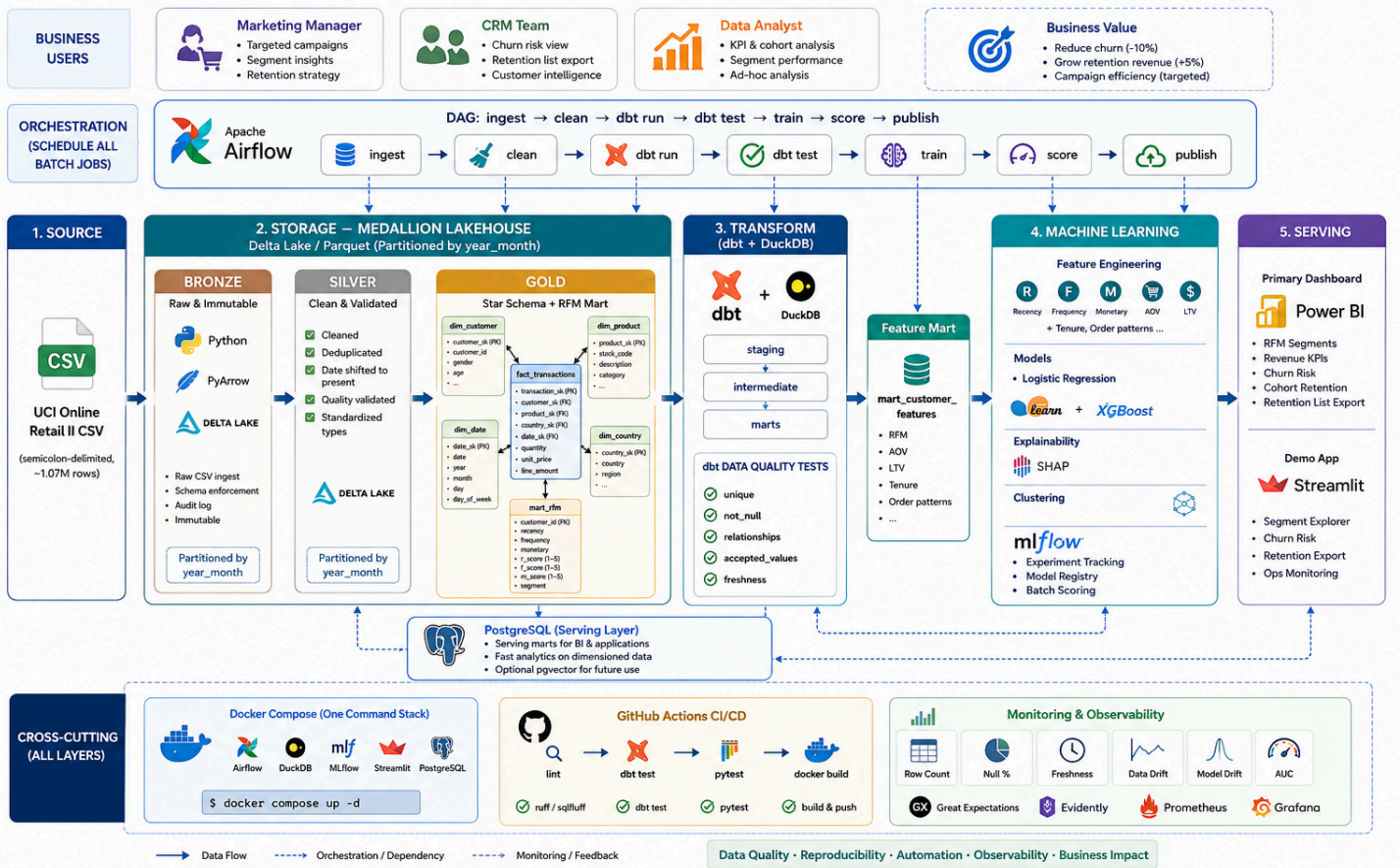
Part B — Solution Architecture

10. Architecture Overview

The platform follows a **Medallion (Bronze → Silver → Gold)** data architecture feeding a **Kimball star schema**, an **RFM + churn** analytics layer, and a serving/dashboard layer. Batch jobs are orchestrated on a schedule; quality is gated; the whole stack is containerised and reproducible.

Retail Customer Intelligence Platform (ACM1)

A 4-Week Mini Enterprise Data Platform • SDLC-First • Dashboard + RFM + Churn Prediction + CI/CD



Design principles

- Business-first, SDLC-complete** — architecture is derived from the BRD, not from technology for its own sake.
- Scope discipline** — deliberately **no Kafka / Spark / Kubernetes / microservices**.
- Immutable raw, governed transforms** — Bronze is append-only; cleaning and modelling are versioned and tested downstream.
- Reproducibility** — one command brings up the full local stack.

10.1 Logical Layers

Layer	Responsibility	Output
Source	The historical transaction dataset (UCI Online Retail II)	online_retail_listing.csv
Bronze	Immutable raw ingest; snake_case, typed, partitioned by year_month; flag cancellations	Parquet partitions
Silver	Clean, dedup, compute line_amount, date-rebase to present, quality gate	Validated Parquet + quality report
Gold	Kimball star schema + RFM mart with quintile scores and segment labels	fact_transactions, dim_*, mart_rfm
Analytics / ML	Feature mart → churn model → batch scores	churn_probability, churn_flag

Layer	Responsibility	Output
Serving	Segments, KPIs, churn-risk, cohort views, retention-list export	Dashboard + exported list

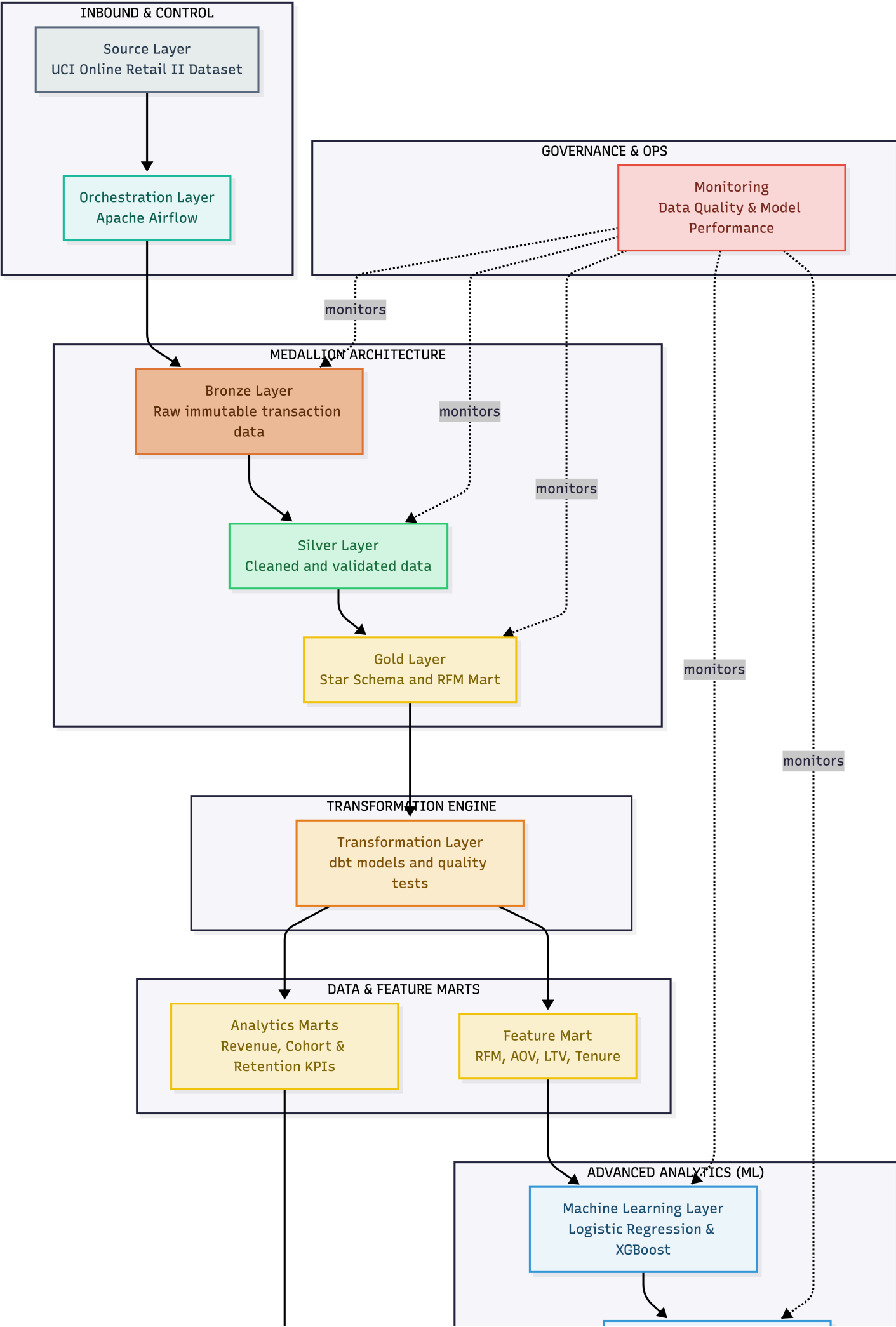
10.2 Cross-Cutting Concerns

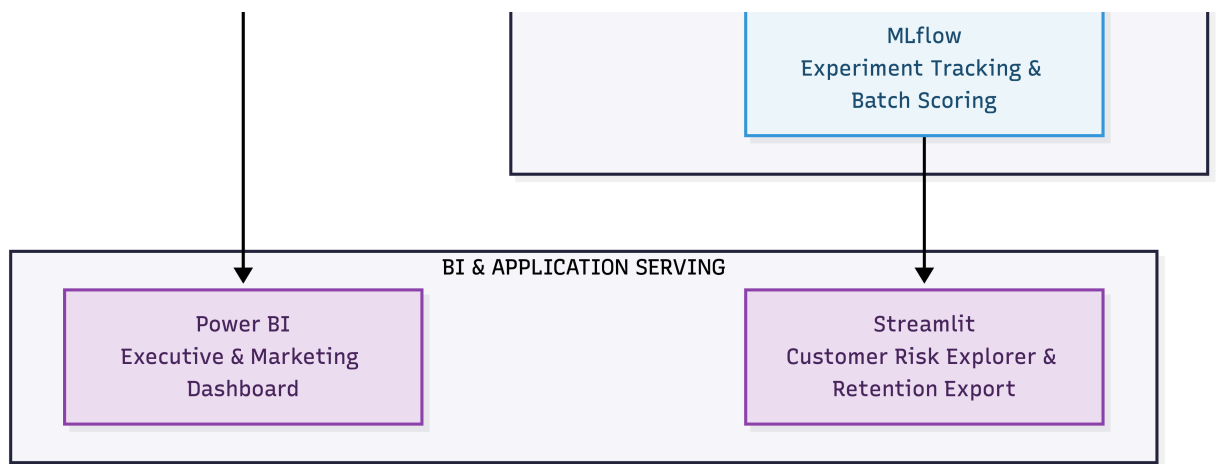
These apply across **all** layers rather than sitting in the data flow:

- **Orchestration (Airflow)** — schedules and sequences every batch stage end-to-end.
- **Containerisation (Docker Compose)** — packages and runs the entire stack consistently.
- **CI/CD (GitHub Actions)** — lint + dbt test + pytest + docker build on every push.
- **Quality & Monitoring** — quality gate (dbt tests) plus per-run observability (data: row counts, null %, freshness; model: prediction drift, AUC, scoring volume).

11. Data Flow

Transform engine target is **dbt + DuckDB** (Silver/Gold as dbt models); Bronze landing stays Python because dbt is not suited to ingesting raw files. The quality gate is implemented as **dbt tests** in CI / the DAG.





12. Dimensional Model (Star Schema)

A central fact table surrounded by conformed dimensions. RFM and churn aggregates are **derived marts** built on top of the fact.

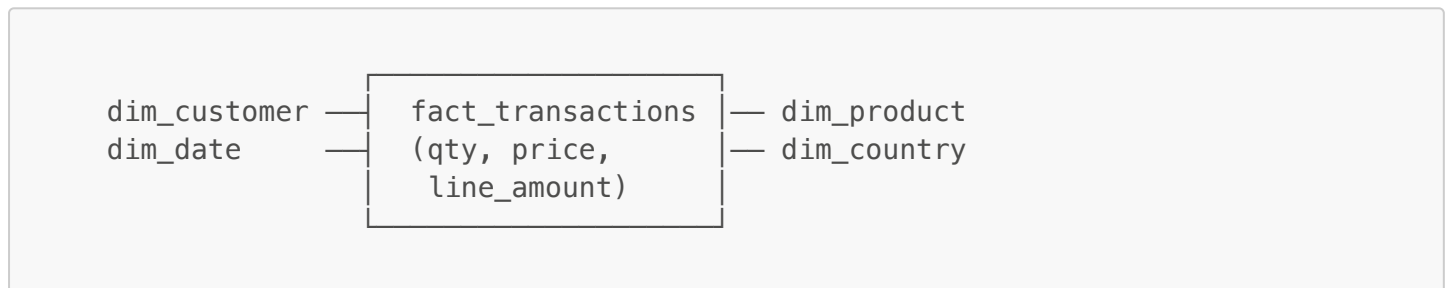


Table	Grain	Key columns
fact_transactions	One row per transaction line	transaction_sk (PK); customer_sk, product_sk, country_sk, date_sk (FK); quantity, price, line_amount
dim_customer	One row per customer	customer_sk (PK), customer_id, first_seen_date, segment
dim_product	One row per product	product_sk (PK), stock_code, description
dim_date	One row per calendar day	date_sk (PK), date, year, month, week, day_of_week, quarter
dim_country	One row per country	country_sk (PK), country_name
mart_rfm	One row per customer	customer_id, recency_days, frequency, monetary, r/f/m_score, segment

Referential integrity note: transactions without a **Customer ID** (~22% of source rows) must be handled explicitly — routed to an "Unknown member" surrogate key or excluded from the fact — so that **fact_transactions.customer_sk** never violates the FK to **dim_customer** (see NFR-1).

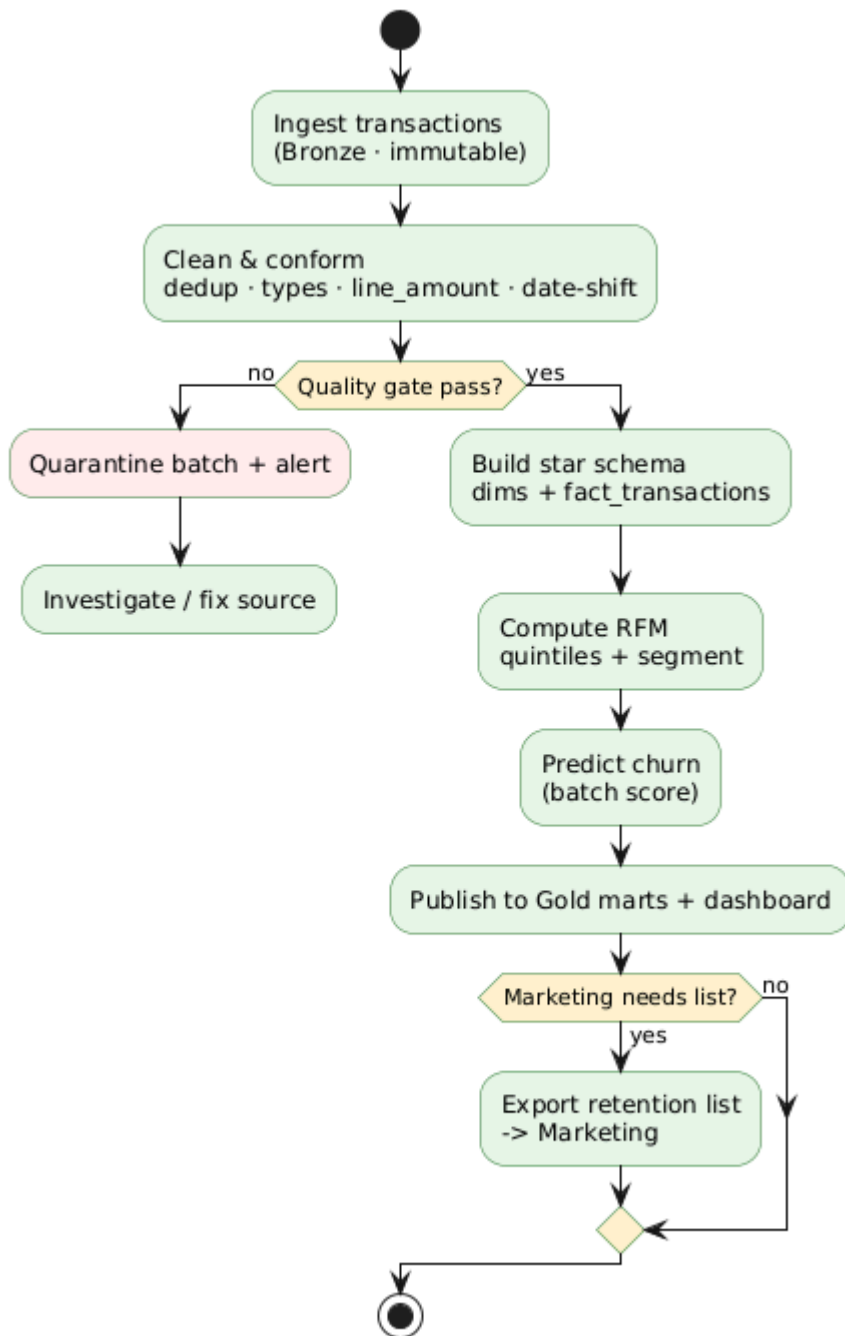
12.1 Technology Stack

Layer	Tool (target)	Responsibility
Orchestration	Apache Airflow	ingest → dbt run → dbt test → train → score → publish
Storage	Parquet (Delta Lake optional)	immutable Bronze, partitioned layers
Transform	dbt + DuckDB	staging → intermediate → marts (star schema + RFM)
ML & tracking	scikit-learn / XGBoost + MLflow	training, experiments, registry, batch scoring
Serving	Power BI (primary) / Streamlit (demo)	segments, KPIs, churn-risk, cohort, retention export
Containerisation	Docker Compose	one-command reproducible stack
CI/CD	GitHub Actions	lint → dbt test → pytest → docker build
Monitoring	metrics table + dashboard "Ops" tab	row counts, null %, freshness, drift, AUC

13. UML — Behaviour & Interaction

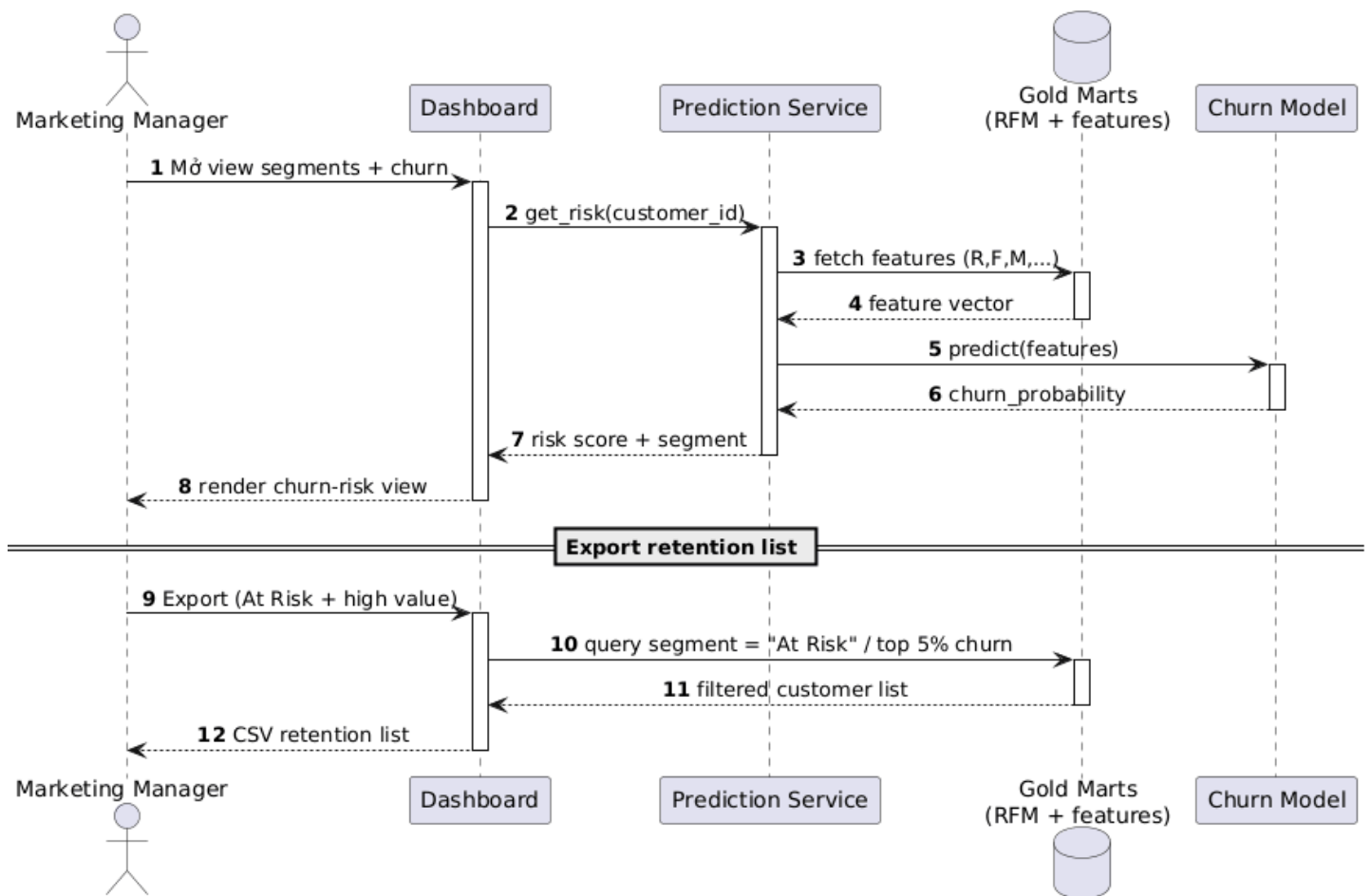
13.1 Activity Diagram — End-to-End Pipeline

The full pipeline with the quality gate / quarantine branch.



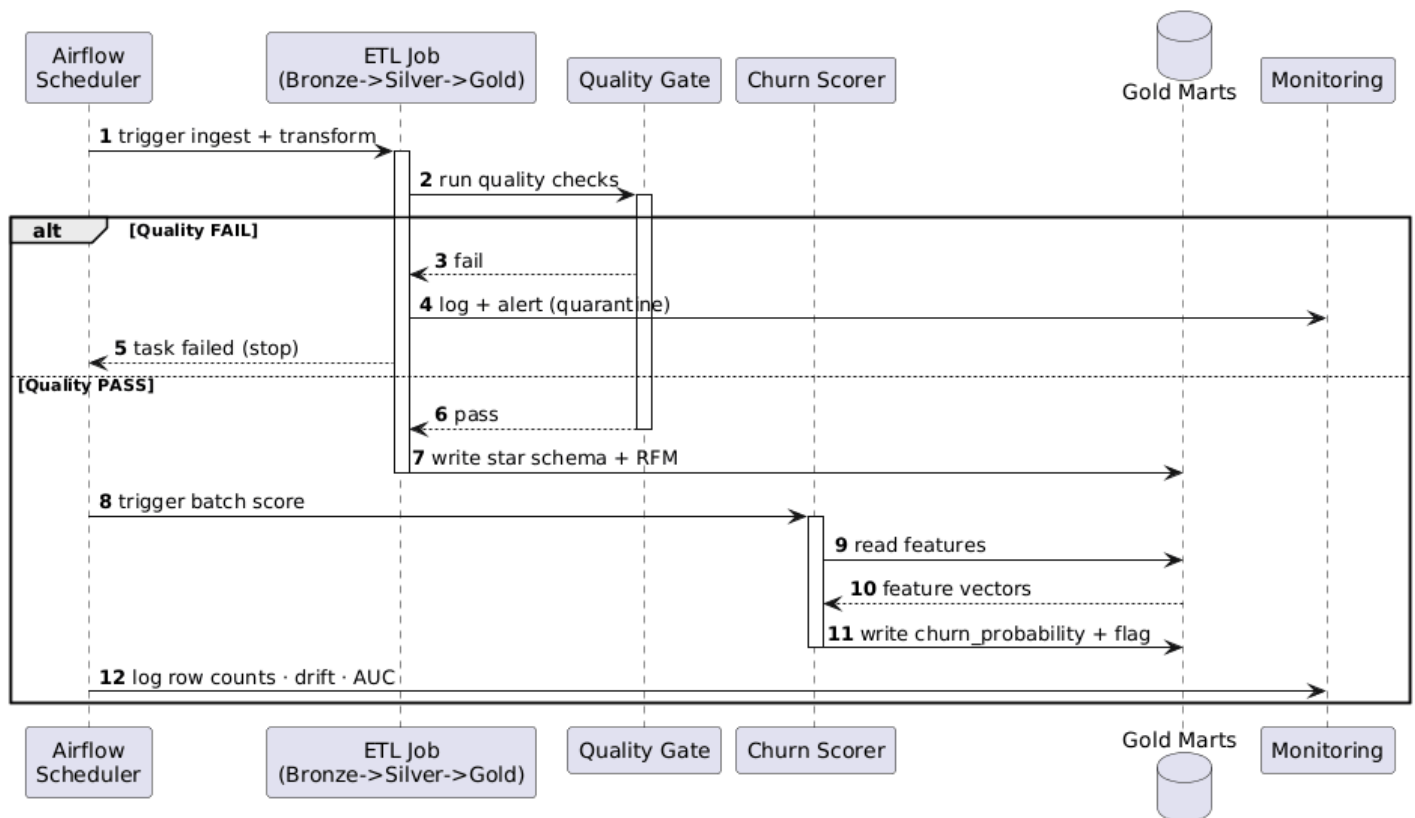
13.2 Sequence Diagram — Dashboard Request

Marketing views churn risk and exports a retention list.



13.3 Sequence Diagram — Scheduled Batch Pipeline

Airflow runs ingest → transform → quality gate → score, with the failure branch.



14. Component → Technology Mapping

Component	Technology
Bronze / Silver / Gold ETL	Python (Bronze) + dbt/DuckDB (Silver/Gold)
Star schema + RFM mart	Gold marts
Data quality checks	dbt tests
Storage	Delta Lake / Parquet
Orchestration	Apache Airflow (scheduled DAG)
Churn model + tracking	LR/XGBoost + MLflow
Dashboard	Power BI / Streamlit (serving layer)
Containerisation	Docker Compose (one-command stack)
CI/CD	GitHub Actions (lint/test/build)
Monitoring	data + model observability

Phased build order (per the Sprint Plan): Star Schema → Airflow → dbt → Dashboard → Docker → Churn → MLflow → CI/CD → Monitoring. The first items earn the "enterprise" judgment; the churn model is intentionally late.

15. Team & Responsibilities

Role	Member	Ownership
Tech Lead	Phan Văn Tiến	BRD · UML · architecture · sprint planning · MLflow · business impact
AI Eng · Data (Team Leader)	Võ Ngọc Gia Bảo	Star schema · dbt models · feature mart · KPI marts · dashboard
AI Eng · Model	Phúc Nhân Nguyễn	RFM logic · feature engineering · churn model · explainability
AI Eng · Pipeline	Ngọc Phương	Repo · Docker Compose · Airflow DAGs · CI/CD
QA · Reviewer	Hoàng Đức Kiên	Definition of Done · test plan · data/model/UAT validation · final review

16. Glossary

Term	Definition
RFM	Recency, Frequency, Monetary — a behavioural scoring method; each scored 1–5 (quintiles) and combined into a segment.
Segment	Business label derived from RFM (e.g. Champions, Loyal, At Risk, Lost).

Term	Definition
Churn	A customer ceasing to purchase within a defined window (rolling 90-day basis for the target metric).
Churn probability / flag	Model output (0–1) and the threshold-applied boolean prediction.
Star schema	Kimball dimensional model: a central fact table surrounded by conformed dimensions.
Medallion (Bronze/Silver/Gold)	Layered data architecture: raw → cleaned/conformed → business-ready marts.
Date-rebasing	Shifting historical timestamps forward by a fixed offset so recency/churn windows align with the present.
Retention list	Exported list of customers (e.g. At-Risk + high-value) for Marketing to action.

17. Related Documents

- docs/ml_design.md — churn model & feature design
- docs/test/test_plan.md — testing strategy & Definition of Done
- docs/planning/Project_Plan.md — 4-week sprint plan
- docs/naming_convention/CONVENTIONS.md — naming & Git workflow
- README.md — repository overview & quickstart

18. Approval / Sign-off

Role	Name	Decision	Date
Tech Lead	Phan Văn Tiến	Approved	02 Jun 2026
QA · Reviewer	Hoàng Đức Kiên	Reviewed	
Team Leader	Võ Ngọc Gia Bảo	Acknowledged	